

Internet of Things:

The industrial segment is currently the fastest growing in the IoT segment

Freescale's 'Internet of Tomorrow' tour in China is tasked with the mission of having over 15,000 customer interactions in the next nine months. In an interview with EFY's Chandramauli Mitra, John Dixon, director - corporate marketing, Freescale, talks about the opportunities that will drive the growth of the IoT



JOHN DIXON
DIRECTOR - CORPORATE
MARKETING, FREESCALE

Q. Which market trends drive the growth of IoT devices?

A. The connected car is the most visual trend. Ten years ago you had 50 components in your car, and in the next ten years you will probably have as many as 500 components. The rate of change in one vehicle space is probably advancing faster than any other industry.

Q. What are the key technologies in microcontrollers (MCUs) that are driving the IoT forward?

A. There are three main MCU technologies. First is reduced power consumption with increased performance. It is important because IoT devices are battery-operated and need to be connected to other devices over a long period of time.

Connectivity comes next, as it consumes more power while connecting to other devices. Therefore there is a growing demand for low-power Bluetooth, ZigBee connectivity.

The third is security, which is the ability to prevent people from tampering with the device as all security documentations are stored in the MCU. To address this issue, Freescale is coming up with solutions like Kinetis that integrate features like encryption and secure boot.

Q. Which key technology trends are making MCUs more secure for IoT applications?

A. Flash and encryption are the two key technological trends that are making MCUs more secure for IoT applications. Security of flash or memory is critical as it stores all the security documentation within it. Data transferred in the network needs to be encrypted for security purposes.

Q. What key challenges do sensors need to address in the future to provide better solutions for IoT applications?

A. Power consumption is one challenge that IoT devices need to address. Embedded sensors exhaust the power of the device, thereby reducing long hours of operation.

The other constraint is the size of the sensors. As the device size shrinks, it becomes extremely difficult to embed more and more sensors in the reduced space.

Q. IoT applications demand low-power components. What changes are being made to make these better?

A. One thing that you can do to reduce power consumption and save space is integrate multiple devices into one single chip, so that you do not have to go from chip-to-chip

for communicating with processors, power management integrated circuits, memory or any other component.

Freescale has released SCM, a technology that allows integration of processors, power integrated circuits, flash and external memory as well as 102 other components like resistors and capacitors all on a single-piece silicon in die form.

The second is process technology. When everything becomes small, it takes less power. A good example is KLO2 or KLO3, which is a Kinetis based device from Freescale. It measures 1.8mm x 1.8mm and can deliver a 48MHz performance.

Q. How do you see the IoT market positioned in Asia when compared globally?

A. In the USA, there is a lot of focus on economic efficiencies related to the IoT, which means, a lot of IoT devices are used in applications that are purely revenue earners. For instance, approximately US\$ 1.8 billion per year is collected from the parking meters installed in San Francisco and New York.

On the other hand, in Asia, the IoT is exercised more for personal or individual use. This could be because of the different pace and focus of the government and consumers.

Q. Which is the fastest growing IoT segment?

A. The industrial segment is currently the fastest growing in the IoT segment. This is the area where money can be saved through efficiencies using the IoT.

Q. What application segments drive the need for the IoT?

A. Applications like parking assistance and advanced driver assistance systems in the automotive segment are among the top application segments.

Apart from industrial automation, applications related to home automation will also witness changes related to the use of the IoT. Compatibility can be achieved through communication protocols like Thread that enable multiple wireless standards to communicate off-the-shelf with other devices.

In smartcities, LED lighting exhibits a huge opportunity and is the easiest to implement.

Q. How do you see India?

A. I see India as an IoT enabler in different forms. India is well-known for its software expertise and innovation, but the biggest challenge the IoT faces here is on software side. ●